



Figure 6: Key services

Host	Status	Duration	Attempt	Last Check	Status Information
localhost	Up	420d 8h 29m 33s	1/10	2011-03-12 19:32:56	OK-127.0.0.1:rita 0.023ms, lost 0%
www.google.com	Up	201d 7h 28m 23s	1/5	2011-03-12 19:34:08	HTTP OK HTTP/1.0 200 OK-9726 bytes in 0.149 seconds
www.nagios.com	Up	N/A	1/5	2011-03-12 19:33:58	HTTP OK HTTP/1.1 200 OK-35616 bytes in 0.698 seconds
www.yahoo.com	Up	201d 7h 24m 58s	1/5	2011-03-12 19:30:45	HTTP OK HTTP/1.0 200 OK-170615 bytes in 1.041 seconds

Figure 7: Host status

monitoring during an entire day, like CPU usage, Nagios host service and Nagios HTTP services. The X-axis shows the time in hours and the Y-axis shows pre-defined states in percentage. The graph is shown in Figure 6.

The blue graph denotes the CPU usage during the entire day, as we can see that according to the time on the X-axis the states are changing during the day. We have pre-defined the levels of critical and warning states as 10 per cent (critical) and 20 per cent (warning state).

The yellow graph shows the packets lost during the entire day. As we can see, the packets are lost only during the first four hours; thus NagiosXI will notify the user every 30 minutes if the critical state is reached.

The red graph shows the round trip average (RPA) time taken by the host to check the plugins performance of the network.

Monitoring website URLs

We can monitor any remote URL by the same process we used to monitor our local checks. We first have to configure the website URL settings in NagiosXI by going to the configuration wizard. Then we can add any URL of the website we want to monitor.

After we have entered the URL, we can change the host name and choose the port number, URL options as well as the URL services. We can setup www.google.com and www.yahoo.com for monitoring.

In Figure 7 we can see the current host status of the Google and Yahoo websites. It shows that there is no error and the attempt count is less than the maximum attempt.

Use the system configuration files and denote the proper IP to achieve the expected target statistics. Sensors like netflow v9 won't work for public networks because we can't access the router.

To summarise, when Nagios is properly configured and installed in the network, it can monitor the system effectively for network usage, processes, activities, as well as CPU, memory and disk usage. Choose the appropriate parameter, as all the inherent parameters to report the system statistics are preinstalled. The user can specify the IP address that requires critical monitoring of inbound and outbound traffic. 

References

- [1] Shival Dubey, Anant Wadhwa, Subodh Kumar, Vineet Virmani and Vaibhav Hans: 'Agent-Less Hardware Monitoring Tool for Cluster Management', International Journal of Advanced Research in Computer Science and Software Engineering, Vol.3, No. 9, September 2013.
- [2] RuanEllis: HOWTO: Set up NAGIOS for an HPC cluster. December 2014. Available from: <https://github.com/alces-software/symphony/wiki/HOWTO-Setup-NAGIOS-for-an-HPC-cluster>
- [3] Nagios report: 'Nagios XI - Installing The Windows Agent: NSClient++' Available from: <https://assets.nagios.com/downloads/nagiosxi/docs/Installing-The-Windows-Agent-NSClient++-for-Nagios-XI.pdf>
- [4] Ravi Saive: 'How to Add a Windows Host to a Nagios Monitoring Server' November 2013. Available from: <http://www.tecmint.com/how-to-add-windows-host-to-nagios-monitoring-server/>
- [5] Magesh Maruthamuthu: 'How to add a Remote Windows Host on a Nagios Server to Monitor', February 2015. Available from: <http://www.2daygeek.com/add-remote-windows-host-on-nagios-server-to-monitor/#>

By: Kiruthika Devi B.S., Abhishek Singh, Karan Malik and Subbulakshmi T.

Kiruthika is a full time Ph.D scholar at Vellore Institute of Technology, Chennai doing her research in security. She can be contacted at kiruthikadevi.bs2015@vit.ac.in
Dr T. Subbulakshmi is a FOSS enthusiast and working as a professor at Vellore Institute of Technology, Chennai. Her research includes FOSS for security. She can be contacted at research.subbulakshmi@gmail.com.

Abhishek Singh and Karan Malik are IV year B.Tech students at Vellore Institute of Technology, Chennai. They can be contacted at abhishek.kumar2013@vit.ac.in and karan.malik2013@vit.ac.in, respectively.